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## Adverse pregnancy outcomes in *Toxoplasma gondii* seropositive Hispanic women

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### Abstract

**Aims**—Chronic *Toxoplasma gondii* infection is not thought to affect pregnancy or birth outcomes, but there are few prospective studies. The study aims were *T. gondii* Immunoglobulin G measurement and relationship of chronic *T. gondii* infection with gestational age at birth and adverse pregnancy outcomes in 690 Hispanic women in Tampa, Florida.

**Methods**—Hispanic women, born either in the U.S. or in Latin America or the Caribbean had a venous blood sample drawn to measure *T. gondii* IgG and *T. gondii* serotype at the first prenatal visit, along with collection of demographic and health related measures. Sero-seropositive and seronegative women were followed through their pregnancy. Gestational age, infant weights, and adverse pregnancy outcomes (miscarriages, preterm births) were compared in the two groups.

**Results**—There were 740 women of self-reported Hispanic ethnicity screened and enrolled in Tampa, Florida, with 690 having birth data extracted from the electronic health record (538 *T. gondii* negative and 152 *T. gondii* seropositive). *T. gondii* seropositivity was 22.4% and the majority (83%) had high avidity titers, indicating chronic infection. Compared to *T. gondii* seronegative Hispanic women, seropositive women had more small for gestational age infants and higher prevalences of miscarriages and preterm birth.

**Conclusions**—This is one of the largest longitudinal cohort studies of women with chronic *T. gondii* infection followed through pregnancy. There was a higher percentages of adverse pregnancy outcomes in this group compared to *T. gondii* seronegative controls. The mechanism for this is unknown and warrants reexamination of the dogma that chronic *T. gondii* infection in pregnant women has no significant clinical consequences.

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## Keywords

chronic infection; pregnancy; preterm birth; miscarriage; *Toxoplasma gondii*

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## Introduction

### *Toxoplasma gondii* Life Cycle and Infection

*Toxoplasma gondii* (*T. gondii*) is a globally distributed intracellular protozoan parasite with seroprevalence estimates of 30% to 50% worldwide infection in humans (1). Highest rates of seropositivity occur in less developed parts of the world and are associated with poor hygiene, contaminated drinking water, and diet. The sexual cycle of the organism occurs in the intestines of Felids, which excrete oocysts that can then be taken up by any warm-blooded vertebrate animal through food and water. The organism can also persist as bradyzoites within tissue cysts. Humans become infected through ingestion of bradyzoites within tissue cysts in undercooked meat or from ingestion of oocysts in water or foods contaminated with oocysts shed in cat feces. After primary infection in the human host, tachyzoites (the rapidly dividing and infectious phase of *T. gondii*) form intracellular cysts in the muscle, brain and eye and transition into bradyzoites, the slow growing, dormant phase of *T. gondii*, which is sequestered from the immune system within the cyst (2). Consequently, a latent infection of toxoplasmosis is thought to persist for the life of the host. Bradyzoites retain some metabolic activity (3) and some parasites may escape from these cysts to provoke a host immune response (4). The acquired immune response in latent toxoplasmosis can be measured by immunoglobulin G (IgG) titers and by avidity assays. Clinically significant reactivation is associated with unchecked replication of tachyzoites, and typically presents as ocular toxoplasmosis in immunocompetent individuals or toxoplasmic encephalitis in immunocompromised hosts with HIV/AIDS or organ transplants (5, 6).

*T. gondii* acute infection acquired by women shortly before or during pregnancy is a major obstetrical concern due to risk for transplacental transmission and congenital infection (congenital toxoplasmosis) (7, 8). It has been a common understanding in obstetrics that chronic, latent *T. gondii* infection does not pose a threat to the mother, the fetus, or the pregnancy, but this clinical dogma has not been examined in large, prospective studies.

The combination of *T. gondii* suppression of immunity in combination with pregnancy related immune changes may act to produce an immunological milieu which could affect the integrity of pregnancy or could promote an immunological state in which reactivation could occur. Reactivations of latent toxoplasmosis have been described in immunocompetent women during pregnancy, although studies are small and cross-sectional (9–12). Pregnancy is associated with immune changes that could theoretically increase the risk of reactivation (shift from Th1 and Th17 balance towards a Th2 response (13), suppression of natural killer (NK) cell axes activation (14)), and upregulated activation markers in monocytes and granulocytes (15, 16)). Regulatory T cells oppose some of these effects along with production of anti-inflammatory cytokines (17). Intracellular *T. gondii* infection-associated immunosuppression signals include blockade in the nuclear factor kappa-B (NF- $\kappa$ B),

macrophage-signaling pathway causing reduction in IL-12 and TNF-alpha production (18). In chronic infection, there is a continuous balance between the parasite and the host with pro-inflammatory host responses being opposed by signaling from the parasite to dampen host response. During pregnancy, immune changes that normally occur might shift this host cell-parasite relationship. Pregnant Hispanic women are therefore a population of importance for exploring the effects of chronic *T. gondii*.

The prevalence of *T. gondii* infection is 30 to 50% in pregnant women in South America, compared to 11% in the U.S. (19, 20). More than half of the 44 million yearly immigrants to the U.S. are from Latin America and the Caribbean (21) and Hispanic women accounted for nearly 23% of U.S. births in 2013. *T. gondii* strain-specific peptides can be identified using immunoassays to infer the infecting genotype in infected humans (22). There are three primary genotypes of *T. gondii* in North America and Europe; Types I, II, or III, which differ in virulence, neurotropism, and phenotype (23–25). Most North American *T. gondii* patients are infected with Type I or Type II parasites (26). Central and South American strains are genetically diverse with many atypical strains that complicate characterization using immunoassays based on genotype-specific peptides (serotypes). Migrants from Latin America and the Caribbean often are infected with the more virulent Type I and an atypical genotype that cross-reacts with both genotype I and genotype II antigens. In our U.S. study samples, individuals from all Hispanic ethnicities who had been born in the US showed a low prevalence of chronic infection and equal distribution of the 4 archetypal serotypes. In contrast, those women born outside the U.S. had a dominance of the non-reactive serotype. This serotype may represent a divergent, non-archetypal strain and has been identified previously in Hispanic populations in Columbia (27). Adverse birth outcomes (preterm birth, miscarriages, and small for gestational age (SGA) are reported to vary greatly by country of origin for Hispanic populations (28). The original aims of this study were focused on psychosocial factors and potential for reactivation in Hispanic pregnant women in Hillsborough County, Florida. We had the opportunity to include pregnancy outcomes in the study and discovered a correlation of latent toxoplasmosis with adverse pregnancy outcomes of SGA, preterm births and miscarriages

## Methods

The University of South Florida (USF Institutional Review Board approved the study and all women provided informed consent. Pregnant women who self-identified as Hispanic were recruited in two prenatal clinics in Tampa, Florida. Exclusion criteria included age under 18 years old, HIV infection, drug and/or alcohol dependency, autoimmune disease or cancer, steroid medications, BMI<20, plans to terminate the pregnancy, or presence of congenital anomalies. All screened women provided a 15 mL venous blood sample at the initial prenatal visit (before the 16th week of pregnancy). Blood samples were collected in tubes with EDTA as anticoagulant and were kept cold and brought to the lab within 24 hours of collection. An aliquot of blood was centrifuged at 1400 rpm for 10 min at 4°C followed by subsequent storage at -80°C. Plasma levels of IgG titers to *T. gondii* (IU/mL) were measured by the *Toxoplasma gondii* IgG ELISA kit (Cat. No. IB79841, IBL America) following the manufacturer's instructions. The qualitative ELISA cut off was 10 IU/ml. Quantitative titers of IgG were determined at each of the subsequent pregnancy visits for

*T. gondii* seropositive women (up to 4 prenatal and 1 postpartum visits). Over 100 samples were tested for IgM with the ELISA kit from the same manufacturer. *T. gondii* serotypes were determined with a peptide-based ELISA assay adapted from Kong et al. (29) and Shobab et al. (22). Serum samples that were *T. gondii* IgG seropositive were tested for antibody reactivity against GRA6 and GRA7 strain-specific allelic peptide motifs. Each serotyping ELISA contains three *T. gondii* seronegative control sera (*T. gondii* IgG levels of less than 2 IU/mL) for calculation of the threshold values for each serotype, i.e., above which normalized assay values were considered positive. Threshold values were determined for each serotype-specific peptide by averaging the OD values of the three normalized *T. gondii* seronegative samples and adding two standard deviations.

An avidity ELISA was performed on all plasma samples from *T. gondii* seropositive women, using a kit from Euroimmun, following kit instructions, to determine if any primary infections were present in the sample, and to estimate chronic infection. Results were grouped in low avidity, borderline avidity and high avidity.

Total enrollment was 740 Hispanic pregnant women. Fifty women were excluded because they delivered at another hospital or were lost to follow up prior to delivery, leaving 690 women with birth data available for analysis. Of these women, 538 were seronegative for *T. gondii* and 152 were seropositive (22%). Extensive clinical and health data were collected on all screened women at the first prenatal visit using an investigator developed form. We used well-validated Spanish versions of the consent form and all the instruments if participants did not read English. All women completed the Edinburgh Postnatal Depression Scale (EPDS), Perceived Stress Scale (PSS), and Medical Outcome Study (MOS) social support scale at the first study visit. The EPDS is a 10-item screening questionnaire that has been used in many studies of both prenatal and postpartum depression, takes less than 10 minutes to complete, and a score above 11 suggests depressive illness of varying severity (30). The PSS is a 10-item scale used to address a person's perceived control over daily life demands, including what is unpredictable, uncontrollable, and burdensome over the past month (31). Each item is rated on a 5-point scale ranging from never (0) to almost always (4). The scores range from 0–56, with higher scores indicating more perceived stress. Cohen et al. reported Cronbach's alpha of 0.85 in three samples and 0.78 in a general population study (31). The Electronic Health Record (EHR) was later reviewed for pregnancy outcomes after birth in these screened seropositive and seronegative women. Infants delivered at less than 37 weeks of gestation were considered preterm. Small for gestational age infants (SGA) were defined as those infants with birth weights at or less than the tenth percentile for gestational age (GA) by gender according to the United States national reference for fetal growth (32). Miscarriages were deliveries before 20 weeks of gestation. GA was determined by the attendant provider using the woman's last menstrual period and pregnancy ultrasound and recorded in the EHR.

Placentae of a subsample of *T. gondii* positive women were examined for pathology, comparing those who had given preterm birth compared to term births. There were 59 women in this subsample, 38 term births compared to 21 preterm births. Placental weights were available from 129 of the 152 *T. gondii* positive women.

All enrolled women were examined up to 4 times in pregnancy and once in the postpartum for health data and *T. gondii* seropositive women were examined for ocular signs of *T. gondii* reactivation, using a DRS retinoscope camera (ICare, Revenio Group, Finland). A consulting ophthalmologist examined the retinal images. This person was *T. gondii* IgG seropositive and had a high titer of IgG and a borderline seropositive IgM at visit 1, which may have indicated more recent infection. The data were included in the subsequent analyses.

### Statistical Analysis

We computed descriptive statistics to characterize the sample and describe the *T. gondii* status and adverse birth outcomes. For categorical variables, frequency and percentage are reported. For continuous variables, mean and standard deviation are reported. For comparisons between the *T. gondii* seropositive and *T. gondii* seronegative groups, the Chi-Square Test was used for categorical variables and the independent sample t-test for continuous variables. Logistic regression was computed controlling for variables of potential importance in adverse birth outcomes. Statistical significance was determined at  $p < 0.05$  level. All analyses were done in SPSS 28.0.

## Results

### Demographics

Of the 690 pregnant women enrolled, 152 were *T. gondii* IgG seropositive (22% positivity) and 538 were seronegative at the first study visit. Demographic characteristics are shown in Table 1. There were small amounts of missing data for some of the demographics, which are indicated in Table 1. *T. gondii* seropositive women were more likely to have been born outside of the U.S., had less education and income, and were more likely to be unemployed and unmarried than the *T. gondii* seronegative women.

### Avidity, Serotypes, Titers, Eye findings

The mean avidity index (A.I.) was  $82.2 \pm 2.79$ , and range was between 37.4 and 373. The majority (83%) of the women had high A.I. ( $< 60$ ), 23% had borderline A.I. (50–60), and only 3 (1.9%) had low A.I. ( $< 50$ ). This indicates that the women in the study were chronically infected. There was no evidence of acute toxoplasmosis in the sample. Over 100 participant samples were spot checked for IgM and all but one was seronegative.

Among the participants, atypical and non-reactive serotypes were the most common (Table 2), and there were no relationships between serotype and adverse pregnancy outcomes by Chi-Square analysis ( $\chi^2$  (3,  $n=151$ ) = 1.46,  $p = 0.69$ ), or GA by one way ANOVA ( $F$  (3,  $n=151$ ) = 0.71,  $p=0.55$ ). There were no relationships between IgG titer, A.I. and serotype. IgG Titer increased slightly across pregnancy. In the retinal images, only one lesion indicating previous chorioretinitis scarring was found.

### *T. gondii* Seropositivity and Country of Birth

*T. gondii* infection status and prevalence by country of birth for all women screened at their first prenatal visit are presented in Table 3. The seropositivity rate for screened Hispanic

women born in the U.S. (N = 234) was 4.3%. In contrast, the seropositivity rate for those born outside of the U.S. (N = 475) ranged widely, from 9% to 60%, depending on the country and the sample size. A chi square test for independence indicated a statistically significant difference of *T. gondii* serostatus between U.S. vs. Hispanic territories/countries of birth, ( $\chi^2$  (1, n=690)=56.4,  $p < 0.001$ ),  $\phi = 0.286$ .

### Psychosocial Variables by *T. gondii* Serostatus

The mean scores on the EPDS was  $4.88 \pm 4.77$  (SD). The range on this score was 0 to 24. The PSS mean score was  $11.7 \pm 6.8$  (SD) and range was 0 to 35. There were no statistically significant differences on these variables by *T. gondii* status.

### Gestational Age, Preterm Birth and Miscarriages

The mean infant birth weight was 3.27 kgs in the *T. gondii* seronegative compared to 3.16 kgs in the *T. gondii* seropositive group (MWU ( $Z = -2.06$ ,  $p = .04$ )). There was a greater prevalence of SGA infants in the *T. gondii* seropositive women (2.2% for *T. gondii* seronegatives, 6.4% for *T. gondii* seropositives, ( $\chi^2$  (1, n=631) = 6.23,  $p = 0.018$ ,  $\phi = 0.099$ ). There were equal distributions of gender by *T. gondii* serostatus, and more twins (NS) in *T. gondii* seropositive (5.6%) compared to *T. gondii* seronegative (2.8%) women ( $\chi^2$  (1, 740) = 3.42,  $p = 0.18$ ). The range of GA for miscarriages was 6 to 19.6 weeks, with a mean of  $12.6 \pm .347$  (S.D.). There was a significant effect of *T. gondii* serostatus on adverse pregnancy events. This includes occurrences of both miscarriages and preterm births, with *T. gondii* seronegative women having an incidence of 17.5 % adverse pregnancy outcomes (94 of 538 women, 34 miscarriages (6.3%) and 60 (11.2%) preterm births), and *T. gondii* seropositive women having an incidence of 31% (42 of 152 women; 14 miscarriages (9.2%) and 28 preterm births (18.4%) ( $\chi^2$  (1, 690) = 7.73,  $p = 0.008$ ,  $\phi = 0.106$ ). To control for potential confounding factors, a multivariable logistic regression model was fitted for adverse pregnancy outcomes and *T. gondii* serostatus while controlling for a group of selected potential confounding variables. First, a logistic regression model was fitted for the adverse pregnancy outcomes with *T. gondii* serostatus as the only predictor. From this model, the unadjusted odds ratio of *T. gondii* positivity for adverse pregnancy outcomes was estimated to be 1.80 (95% CI 1.186 – 2.74;  $p = 0.006$ ). Three groups of potential confounding variables were then added in a hierarchical manner: (1) Demographics: household income, years of education, employment, marital status, age and country of birth (2) Risk factors for preterm or miscarriage: smoking during pregnancy, hypertension diagnosis, numbers of previous preterm births and miscarriages and EPDS scores above 11, and 3) *T. gondii* IgG titer. Due to the missing data on some of the demographic variables, the sample N for the final model was reduced from 690 to 640 (528 *T. gondii* seronegatives, and 112 *T. gondii* seropositives). Likelihood ratio tests were used to test the significance of these factors on the relationship between *T. gondii* status and adverse pregnancy outcomes. The model that best fit the data included previous preterm pregnancies, hypertension diagnosis, and *T. gondii* seropositive status. Parameter estimates with odds ratios (OR) are indicated in Table 4. As shown in Table 4, after controlling for the listed confounding factors, the highest OR of adverse pregnancy events were predicted by the presence of hypertension in pregnancy, followed by previous history of preterm births. The prediction of adverse pregnancy of those who were *T. gondii* seropositive, when these other factors were controlled was 1.7 times of



the odds of adverse pregnancy events of those who were *T. gondii* seronegative (OR = 1.7, 95%; CI = 1.059–2.726).

SGA was at greater prevalence in the infants of *T. gondii* seropositive (6.4%) compared to seronegative women (2.1%), ( $X^2(1, 641) = 6.86, p = .014, \phi = .102$ ). SGA was more common in twin pregnancies, in mothers who smoked, mothers with preeclampsia or hemorrhage, and mothers who spoke less English. A forward conditional logistic regression on SGA, entering these variables as independent covariates, and other potential variables contributing to SGA (country of birth, age, alcohol use during pregnancy, BMI, stress, depression) was performed using maximum likelihood ratio estimations. *T. gondii* serostatus was entered alone in stage 1, and had an OR of 3.15, 95% CI- 1.28–7.77,  $p = 0.013$ . However, when all the potential covariates were added, the final model excluded *T. gondii* serostatus as preeclampsia, twinning, and smoking during pregnancy had much higher odds ratios than the *T. gondii* effect. The model parameters for this regression are in Table 5.

## Placenta

Placental pathology was not available for the miscarriage cases. Placental weight was significantly lower in the preterm births  $387 \text{ Gms} \pm 134.80$ , compared to term births ( $499.4 \text{ Gms} \pm 115.40$ ,  $t = 3.38, p < .001$ ). The only pathological finding that differed by preterm birth was perivillous fibrin deposition (4 cases out of 21 preterm birth, and 1 case in 38 term births). There were equal and few numbers of chorioamnionitis, chorangitis, and cysts. The pathology summaries indicated greater cord edema in the preterm birth (23%) compared to term births (13.2%). The total number of pathological findings in the preterm births was 13/21 (62%) compared to term birth 18/38 (47%). Lesions included arteritis, villitis, villous agglutination, and chorionic plate vasculitis. There were too few placental analyses available for SGA comparisons.

## Discussion

*T. gondii* is a global parasite, with highest prevalence of seropositive individuals in low income and low development countries (33). The large immigrant population in the U.S. is likely to have higher rates of *T. gondii* infection and require more attention for both surveillance and obstetrical care due to health conditions and previous medical access. In our study those born outside the U.S. had the highest rates of latent toxoplasmosis (mean of 30.4%), compared to reported *T. gondii* infection incidence of 11% in the United States (21). We found no evidence of acute infection or reactivation by retinoscopic examination, IgM levels, or avidity indices. The occurrence of miscarriages and preterm births were higher in the *T. gondii* seropositive women than in *T. gondii* seronegative women and social determinants of health (SDoH) such as socioeconomic status, marital status, and education did not influence this relationship, as well as personal factors such as country of birth, age and BMI. We found that common associations for preterm birth were present in this Hispanic population, such as previous preterm birth and hypertension. The association of previous preterm birth with current adverse pregnancy outcomes is not surprising as one of the major risk factors for preterm birth is previous preterm births, although two thirds of preterm births have unknown etiologies (34). Hypertension is one of the best-known

predictors for preterm birth, as was also apparent in this sample (35). Nevertheless, these factors did not reduce the influence of chronic *T. gondii* infection. Placental pathology did not give any specific clues that have been associated with preterm birth, such as placental insufficiency (36). It is possible that subtle immune differences related to the constant surveillance and suppression of bradyzoites in cysts may be influential in pregnancy when the immune system is so differentially regulated (37). An example is the significantly lower levels of many plasma cytokines in *T. gondii* seropositive pregnant women (38). These effects could potentially involve placental immune function.

The rate of adverse pregnancy outcomes in the *T. gondii* negative sample mirrored national statistics for Hispanic women. The national incidence of preterm birth for Hispanic women in the U.S is 10% (<https://www.cdc.gov/reproductivehealth/maternalinfanthealth/pretermbirth.htm>) and our *T. gondii* negative sample had a 11.2% incidence. The national miscarriage rate is more difficult to determine as many miscarriages occur before a woman knows she is pregnant, but about 10–15% of pregnancies terminate by miscarriage (<https://www.marchofdimes.opubmedrg/complications/miscarriage.aspx>) and our *T. gondii* negative women had an incidence of 6.3%. The rate of SGA in live births in the U.S. for Hispanics is about 6.48% (39) and this sample of *T. gondii* seronegative Hispanic women had an incidence of only 2.1%.

A risk for miscarriage and prematurity in women with chronic *T. gondii* infection has been previously suggested, but the conclusions of these studies have been a matter of dispute. No miscarriages were reported in chronically infected individuals in an Egyptian study (40). Other studies reported low frequencies of miscarriage in pregnant females with primary, acute infection: 3% among pregnant females in Italy (41), 6% among pregnant females in Norway (42), and 10% among Finnish women (43). Among pregnant Egyptian females, there was an increased incidence of miscarriage, ranging from 14–30%, among females considered to have primary infection, but no miscarriages were reported in chronically infected individuals (40). In Bangladesh, one-third of pregnant females with unspecified acute or chronic *T. gondii* infection experienced miscarriage compared to one-quarter of non-infected pregnant females (44). A study conducted in Norway identified a trend toward increased risk for miscarriage between the 16th and 19th week of gestation in pregnant females with chronic infection after stratifying by GA (45). Li and colleagues (46) conducted a meta-analysis and found that odds of miscarriage were significantly higher in females with primary maternal infection compared to non-infected pregnant females. A retrospective study of 850 cord blood samples from Iranian women, presumed to have latent infection, found that odds of having a miscarriage history was two and a half times higher in *T. gondii* IgG seropositive women (OR = 2.45, CI = 1.56–3.87,  $P < 0.001$ ) (47). This paper concluded that chronic *T. gondii* infection may be one of the major worldwide causes of miscarriage. A recent report of chronically infected women with high avidity infection showed that parasite DNA was detectable in blood and placental samples from spontaneous abortions, suggesting that congenital infection was a possibility in these women with chronic toxoplasmosis. (48). A recent systematic review and meta-analysis reported that recent *T. gondii* infection is associated with an elevated risk of spontaneous abortion, confirming our results (49).



Miscarriages appear to be the most well studied adverse pregnancy outcome in the previous research, and preterm birth has not been well characterized in women with latent toxoplasmosis. Preterm birth has been observed to be associated with acute *T. gondii* infection (46), but a retrospective study in Brazil found no significant difference in the incidence of preterm birth when comparing acutely infected or chronically infected pregnant females to seronegative pregnant females (50). In the current study, preterm birth was a more prevalent finding than miscarriage, and both were higher in *T. gondii* seropositive women compared to seronegative controls.

There was also a higher incidence of SGA in *T. gondii* seropositive women in the study but this effect was diminished in the logistic regression final model by the very significant effects of preeclampsia, twin pregnancy, and smoking, and the contribution of multiple other factors that may be important in infant prenatal growth. SGA is associated with early and later life morbidities such as cerebral palsy and phenotype such as short stature (51). There are multiple factors that may contribute to SGA, including genetic and epigenetic influences, placental insufficiency, and maternal lifestyle factors such as smoking (52). Our results indicate that chronic *T. gondii* infection may also contribute to SGA and deserves more focused research.

There was no evidence of reactivation during pregnancy by retinal lesions or symptomatology. However, reactivation is not required for a number of reported biological and behavioral manifestations of chronic toxoplasmosis, including lack of aversion to predators in mice (53), schizophrenia in humans (54), and obesity (55) in people who are chronically infected. Sustained epigenetic effects have been proposed as a possible mechanism for these long term consequences of toxoplasmosis (56), but the causal relationship between *T. gondii* chronic and behavior remains controversial.

Throughout the lifelong course of chronic infection there are consistently high titers of *T. gondii*-specific IgG and altered immune responses including cytokine levels and functions (57). This immune response itself may be associated with the adverse birth outcomes we have reported even in the absence of parasite recrudescence. The immune changes elicited by either acute or chronic toxoplasmosis may provoke adverse pregnancy outcomes as a downstream result of immune-initiated pathways. Chronic *T. gondii* infection has been studied largely in the brain, but these parasites can be found in other sites (heart and skeletal muscle) and can be even transmitted to transplant recipients via organ transplantation (58, 59). Much less is known about chronic infection in other tissues, and whether there are effects of chronic infection in the female reproductive tract or in pregnancy remains unknown. Alternatively, *T. gondii* seropositivity may be a biomarker of an immunological state that predisposes to poor pregnancy outcomes. Chronic infection in pregnancy is known to be associated with changes in cytokines and chemokines (60) (61). Bradyzoites have a large number of genes expressed during the chronic stage, which promote the chronicity of the infection by blocking transcription of inflammatory cytokine genes (62).

Regardless of the mechanisms, our findings suggest that evidence of chronic infections such as toxoplasmosis may influence obstetric outcomes. If so, clinicians should be alerted and aware of these possible risks. Considering the enormous global burden of adverse birth

outcomes for both mothers and infants, further investigations into associations and causality of poor obstetric outcomes with latent toxoplasmosis are justified. Efforts such as provision of clean water and avoidance of uncooked or undercooked meat should be increased to lower the rate of *Toxoplasma* infections particularly in areas of the world with high prevalence.

### Limitations

The study was not initially designed to focus on obstetrical complication or prematurity, so some important variables were not routinely collected. Further research to explore mechanisms needs to include more precise risks and measures associated with miscarriage, preterm birth and SGA in women with chronic toxoplasmosis. This sample may have had limited previous health care in their countries of origin, have suffered multiple other infectious disease, and had nutritional deficits that could be playing a role in the current findings. They were all Hispanic, and many were immigrants. They were generally economically and sociologically disadvantaged. A larger and more heterogeneous population should be studied in the future.

A limitation is that no fetal chromosomal testing or placental pathology was done in the miscarriage cases, so clues to etiology cannot be discovered.

The occurrence of the adverse pregnancy outcomes in chronic Toxoplasmosis provides an unusual model for chronic infection during pregnancy, as we found no evidence for acute toxoplasmosis. With such a large sample size, discrete immune, endocrine and metabolic data was difficult to acquire. We have a subsample currently under investigation for many of the biological measures that may shed light on our findings.

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### Data Availability Statement:

Deidentified data are available by request of the primary author. No data have been deposited in repositories.

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**Table 1**Demographic Data of 704 screened women by *T. gondii* status.

| Demographic variable | <i>T. gondii</i> negative N=552 | <i>T. gondii</i> positive N=152 | Chi Square or t-test       |
|----------------------|---------------------------------|---------------------------------|----------------------------|
| Education            |                                 |                                 |                            |
| Grammar              | 14 (2.6%)                       | 3 (2.0%)                        | $\chi^2=23.1$ , $p<0.001$  |
| Middle school        | 97 (18%)                        | 52 (34.7%)                      |                            |
| High school          | 336 (62.5%)                     | 66 (44%)                        |                            |
| College              | 76 (14.1%)                      | 26 (17.3%)                      |                            |
| Postgraduate         | 15 (2.8%)                       | 3 (2.0%)                        |                            |
| Missing              | 14                              | 2                               |                            |
| Income               |                                 |                                 |                            |
| <\$4,999             | 105 (20%)                       | 31 (22.8%)                      | $\chi^2=16.9$ $p<0.005$    |
| \$5000-\$14,999      | 111 (21.1%)                     | 39 (26%)                        |                            |
| \$15,000-\$24,999    | 154 (29.3%)                     | 58 (38.7%)                      |                            |
| \$25,000-\$39,999    | 108 (20.5%)                     | 15 (10.1%)                      |                            |
| \$40,000-\$69,999    | 45 (8.6%)                       | 5 (3.3%)                        |                            |
| >\$70,000            | 3 (0.6%)                        | 2 (1.3%)                        |                            |
| No Information       | 12                              | 2                               |                            |
| Employment           |                                 |                                 |                            |
| Part time            | 84 (15.7 %)                     | 21 (14.8%)                      | $\chi^2=8.86$ , $p<.012$   |
| Full time            | 155 (29%)                       | 27 (18%)                        |                            |
| Unemployed           | 295 (55.2%)                     | 102 (69%)                       |                            |
| No Information       | 4                               | 2                               |                            |
| Marital status       |                                 |                                 |                            |
| Single               | 332 (61.7%)                     | 82 (52.0%)                      | $\chi^2=12.95$ , $p<0.002$ |
| Married              | 186 (34.6%)                     | 56 (37.3%)                      |                            |
| Divorced/Separated   | 20 (3.7%)                       | 16 (10.7%)                      |                            |
| No Information       | 13                              | 2                               |                            |
| Ethnicity            |                                 |                                 | NS                         |
| Unknown              | 9 (1.7%)                        | 0 (0%)                          |                            |
| Hispanic White       | 479 (89%)                       | 139 (92.7%)                     |                            |
| Hispanic Black       | 19 (3.5%)                       | 7 (4.7%)                        |                            |
| Other Non-White      | 31 (5.8%)                       | 4 (2.7%)                        |                            |
| No Information       | 0                               | 2                               |                            |
| Age (Mean Years)     | 28.6 $\pm$ 6.0                  | 32.2 $\pm$ 6.3                  | $t=-6.5$ , $p<0.001$       |
| BMI                  | 30.6 $\pm$ 7.1                  | 30.6 $\pm$ 6.7                  | $t=-.02$ , $p=.98$         |
| Birth Country        |                                 |                                 |                            |
| USA born             | 208 (38.6%)                     | 10 (6.5%)                       | $\chi^2=56.4$ , $p<0.001$  |
| Non USA born         | 330 (61.3%)                     | 142 (93.4%)                     |                            |

*T. gondii* positive Hispanic women compared to *T. gondii* negative Hispanic women were more likely to be older, born outside the U.S., less educated, unemployed, have lower income and be divorced or separated.



**Table 2**

Serotype Frequencies in T.gondii positive women

|              | Frequencies (%) for Term births (N=108) | Frequencies (%) for births before 37 weeks (N=44) |
|--------------|-----------------------------------------|---------------------------------------------------|
| I/III        | 17.6%                                   | 23.8%                                             |
| II           | 17.6%                                   | 20.8%                                             |
| Atypical     | 25.0%                                   | 28.6%                                             |
| Non-reactive | 39.8%                                   | 35.7%                                             |

The distribution of major serotypes did not differ significantly by preterm compared to term birth

**Table 3**

T. gondii prevalence by country of birth

| Country of Birth                                                        | T.gondii negative N=538 | T.gondii positive N=152 | Prevalence (% positive) |
|-------------------------------------------------------------------------|-------------------------|-------------------------|-------------------------|
| United States                                                           | 208                     | 10                      | 4.6                     |
| Mexico                                                                  | 150                     | 55                      | 26.8                    |
| Puerto Rico                                                             | 74                      | 7                       | 8.6                     |
| Honduras                                                                | 16                      | 22                      | 57.9                    |
| Cuba                                                                    | 43                      | 21                      | 32.8                    |
| Dominican Republic                                                      | 11                      | 8                       | 42.1                    |
| Venezuela                                                               | 5                       | 2                       | 28.6                    |
| Columbia                                                                | 9                       | 4                       | 30.8                    |
| <i>Others (Guatemala, El Salvador, Brazil, Peru, Panama, Nicaragua)</i> | 22                      | 23                      | 51.1                    |

The prevalence of T. gondii seropositivity varied greatly by country/territory of birth, with the U.S. being lowest.

**Table 4**

## Logistic Regression Analysis of Adverse Pregnancy Outcomes

| Variables                           | B      | S.E.  | Wald   | df | Sig.  | Exp(B) | 95% C.I for EXP(B) |       |
|-------------------------------------|--------|-------|--------|----|-------|--------|--------------------|-------|
|                                     |        |       |        |    |       |        | Lower              | Upper |
| Education                           | −0.062 | 0.153 | 0.167  | 1  | 0.683 | 0.939  | 0.696              | 1.268 |
| Household Income                    | 0.111  | 0.103 | 1.160  | 1  | 0.281 | 1.117  | 0.913              | 1.367 |
| Marital Status (Single)             |        |       | 1.019  | 2  | 0.601 |        |                    |       |
| Marital Status (Married)            | −0.259 | 0.257 | 1.018  | 1  | 0.313 | 0.772  | 0.467              | 1.276 |
| Marital Status (Divorced/Separated) | −0.104 | 0.475 | 0.048  | 1  | 0.827 | 0.901  | 0.355              | 2.289 |
| Employment (Part Time)              |        |       | 2.125  | 2  | 0.346 |        |                    |       |
| Employment (Full Time)              | −0.226 | 0.352 | 0.412  | 1  | 0.521 | 0.798  | 0.401              | 1.590 |
| Employment (Unemployed)             | −0.451 | 0.320 | 1.987  | 1  | 0.159 | 0.637  | 0.340              | 1.193 |
| Years Living in the US              | −0.009 | 0.012 | 0.596  | 1  | 0.440 | 0.991  | 0.968              | 1.014 |
| Current Depression                  | 0.457  | 0.275 | 2.758  | 1  | 0.097 | 1.579  | 0.921              | 2.708 |
| Previous Miscarriages               | 0.133  | 0.239 | 0.312  | 1  | 0.577 | 1.143  | 0.715              | 1.825 |
| Previous Preterm Birth              | 1.192  | 0.268 | 19.843 | 1  | 0.000 | 3.293  | 1.949              | 5.564 |
| Smoked in Pregnancy                 | 0.583  | 0.456 | 1.636  | 1  | 0.201 | 1.791  | 0.733              | 4.377 |
| Hypertension                        | 1.133  | 0.285 | 15.804 | 1  | 0.000 | 3.106  | 1.776              | 5.432 |
| <i>T.gondii</i> Positive            | 0.590  | 0.258 | 5.219  | 1  | 0.022 | 1.805  | 1.088              | 2.995 |
| Intercept                           | −1.860 | 0.587 | 10.047 | 1  | 0.002 | 0.156  |                    |       |

**Variables entered:** Household income (6 levels), marital status (single, married, divorced/separated), education, employment status (Full time, Part time, Unemployed), age, years of living in the US, EPDS score of 10 or higher, smoking during pregnancy, number of previous preterm births and miscarriages, *T.gondii* positive status.  $\beta$  = estimated effect; S.E. = standard error of  $\beta$ ; Wald = the Wald test statistic; df= degree of freedom; Exp ( $\beta$ ) = Odds Ratio (OR); and 95%CI = 95% confidence interval for the OR.

**Table 5**

## Variables in the Equation

|                     |              |       |      |       |      |      |        | 95% C.I for EXP(B) |        |
|---------------------|--------------|-------|------|-------|------|------|--------|--------------------|--------|
|                     |              | B     | S.E. | Wald  | df   | Sig. | Exp(B) | Lower              | Upper  |
| Step 1 <sup>a</sup> | Preeclampsia | 2.84  | 0.87 | 10.71 | 1.00 | 0.00 | 17.17  | 3.13               | 94.24  |
|                     | Constant     | -4.23 | 0.58 | 52.89 | 1.00 | 0.00 | 0.01   |                    |        |
| Step 2 <sup>b</sup> | Twins        | 2.99  | 1.12 | 7.18  | 1.00 | 0.01 | 19.88  | 2.23               | 176.94 |
|                     | Preeclampsia | 3.04  | 0.98 | 9.72  | 1.00 | 0.00 | 20.89  | 3.09               | 141.28 |
|                     | Constant     | -4.71 | 0.72 | 42.33 | 1.00 | 0.00 | 0.01   |                    |        |
| Step 3 <sup>c</sup> | Twins        | 3.61  | 1.33 | 7.34  | 1.00 | 0.01 | 36.81  | 2.71               | 499.58 |
|                     | Preeclampsia | 3.69  | 1.21 | 9.37  | 1.00 | 0.00 | 40.14  | 3.77               | 427.08 |
|                     | Smoking      | 3.80  | 1.50 | 6.45  | 1.00 | 0.01 | 44.70  | 2.38               | 840.04 |
|                     | Constant     | -5.41 | 1.02 | 28.13 | 1.00 | 0.00 | 0.00   |                    |        |

<sup>a</sup>.Variable(s) entered on step 1: Preeclampsia

<sup>b</sup>.Variable(s) entered on step 2: Twins

<sup>c</sup>.Variable(s) entered on step 3: Smoking

Model of SGA infants after T. gondii status, hemorrhage, alcohol, infant gender, country of birth, preferred language, stress and BMI removed.  $\beta$  = estimated effect; S.E. = standard error of  $\beta$ ; Wald = the Wald test statistic; df= degree of freedom; Exp ( $\beta$ ) = Odds Ratio (OR); and 95%CI = 95% confidence interval for the OR.